

Roll No. .

Total No. of Questions : 06]

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EYS-166

B. Tech. (New Scheme) Examination, 2024

(Fourth Semester)

MICROPROCESSOR AND

MICROCONTROLLER

EL-405

Time : 3 Hours]

[Maximum Marks : 60

Note : Q. No. 1 is compulsory. Attempt any *two* parts from remaining questions.

1. Write short answers of the following : **5×2=10**
 - (a) Name the flags of 8086 and give their functions.
 - (b) List different data transfer schemes used in 8085.
 - (c) Give advanced features of 80386.
 - (d) What is the function of IC8212 ?
 - (e) Name SFRs used in 8051 for serial communication and give their function.

2. (a) Discuss the function of the following Pins 8085.
 - (i) HOLD & HLDA
 - (ii) INTR & INTA'
 - (iii) SID & SOD
 - (iv) X1 & X2
 - (b) Interface three memories of size 2KX8, 4KX8 and 8KX8 to 8085 and give their address range.
 - (c) Write a program for 8085 to find the largest number from the given series of 10 numbers stored in consecutive memory locations. 10
3. (a) Discuss the Internal architecture of 8086.
 - (b) Describe minimum and maximum mode of operation of 8086.
 - (c) Explain real and protected mode of operation used in memory Management. 10
4. (a) Discuss the advancement in Intel series processors (8086 onwards).
 - (b) Describe the super Scalar architecture used in Pentium Processors.
 - (c) Write a short note on SPARC Processors. 10

5. (a) Discuss the operation of IC 8255 with the help of its control word. Explain different modes of operation.
 - (b) Draw the block diagram of IC8259 and explain its operation.
 - (c) Write a short note on IC 8251. 10
6. (a) Describe the External memory Interfacing in IC 8051.
 - (b) List different addressing modes available in IC 8051 with suitable example.
 - (c) Explain, how Timer Operation takes place in IC8051 ? 10